

CS231n 1-3

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K-NN의 학습과 테스트

Train 방법

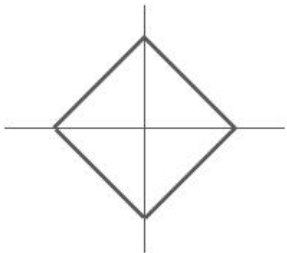
모든 학습데이터를 저장한다

Test 방법

Train데이터와 test데이터를 계산하여 가장 가까운 K개의 train데이터를 찾아 label을 결정한다

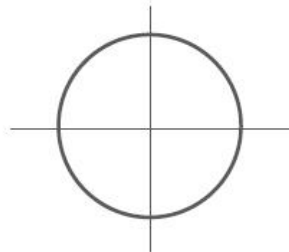
L1 (Manhattan) distance

$$d_1(I_1, I_2) = \sum_p |I_1^p - I_2^p|$$



L2 (Euclidean) distance

$$d_2(I_1, I_2) = \sqrt{\sum_p (I_1^p - I_2^p)^2}$$



L1 distance: $d_1(I_1, I_2) = \sum_p |I_1^p - I_2^p|$

test image				training image				pixel-wise absolute value differences			
56	32	10	18	10	20	24	17	46	12	14	1
90	23	128	133	8	10	89	100	82	13	39	33
24	26	178	200	12	16	178	170	12	10	0	30
2	0	255	220	4	32	233	112	2	32	22	108

=

add → 456

K-NN의 학습과 테스트

KNN 단점

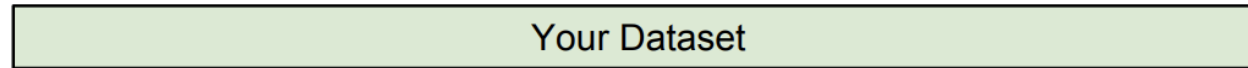
- 일반적인 모델과 달리 train과정이 짧고 prediction과정이 오래 걸린다 ($O(N)$)
- 픽셀의 거리계산 값에는 정보가 없다



Hyper Parameter 설정 방법

Idea #1: Choose hyperparameters that work best on the data

BAD: $K = 1$ always works perfectly on training data



Idea #2: Split data into **train** and **test**, choose hyperparameters that work best on test data

BAD: No idea how algorithm will perform on new data

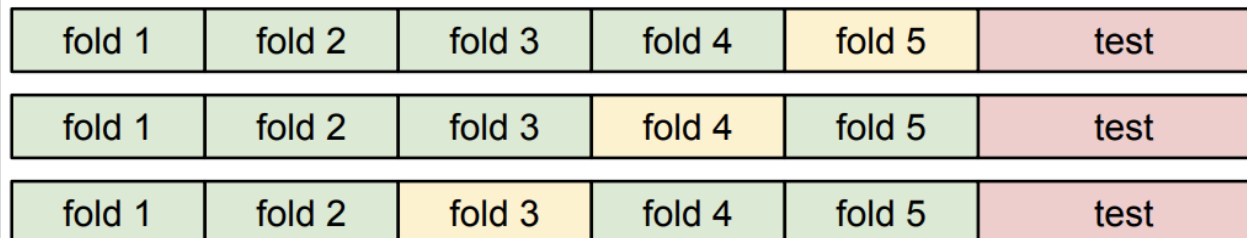


Idea #3: Split data into **train**, **val**, and **test**; choose hyperparameters on val and evaluate on test

Better!



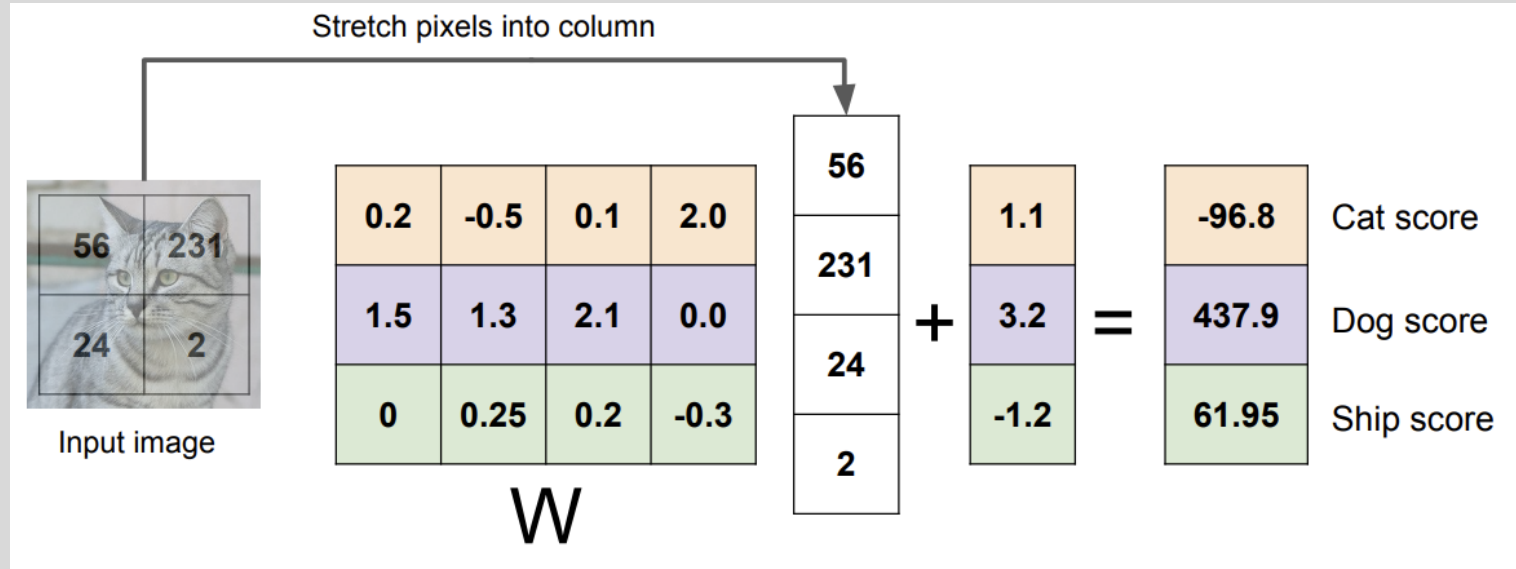
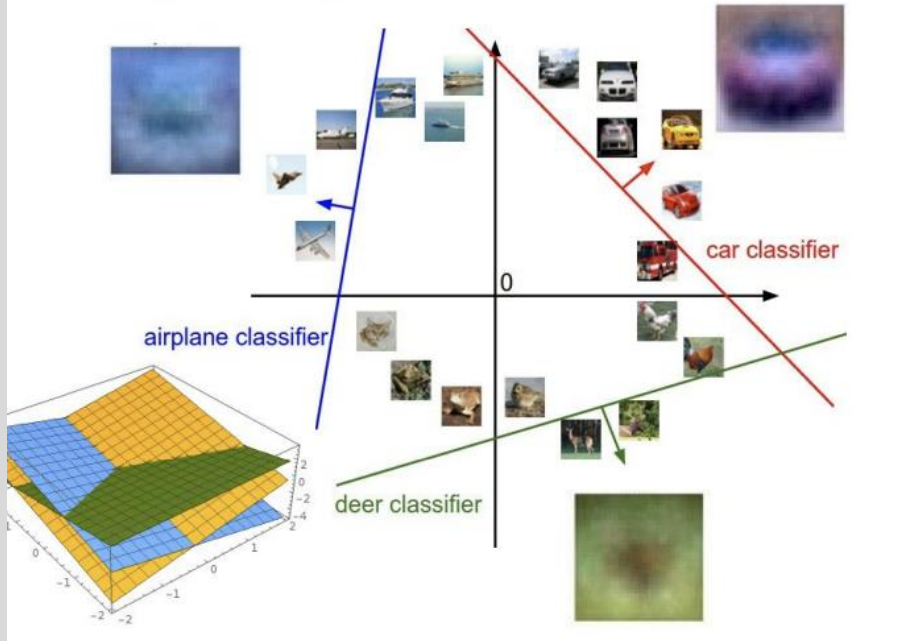
Idea #4: Cross-Validation: Split data into **folds**, try each fold as validation and average the results



Useful for small datasets, but not used too frequently in deep learning

Linear Classifier의 학습과 테스트

Interpreting a Linear Classifier



Score기반의 Loss함수를 계산하여 Loss를 줄이는 방식으로 훈련을 진행

Multiclass loss function

Multiclass SVM loss

Multiclass SVM loss:

Given an example (x_i, y_i)
where x_i is the image and
where y_i is the (integer) label,

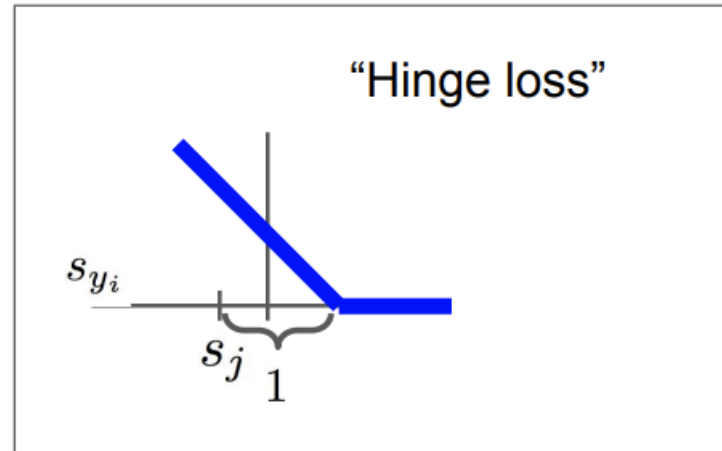
and using the shorthand for the
scores vector: $s = f(x_i, W)$

the SVM loss has the form:

$$L_i = \sum_{j \neq y_i} \begin{cases} 0 & \text{if } s_{y_i} \geq s_j + 1 \\ s_j - s_{y_i} + 1 & \text{otherwise} \end{cases}$$
$$= \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

정확한 클래스의 점수가 틀린 클래스의 점수보다 1이상 크다면 loss는 0이고
정확한 클래스의 점수가 낮다면 틀린 클래스와의 점수 차이 +1 만큼의 loss를
가진다

Multiclass SVM loss:



Multiclass loss function

Regularization

새로운 데이터에 대한 성능이 떨어지는 것을 방지하기 위해 Regularization 을 적용한다

Regularization

λ = regularization strength
(hyperparameter)

$$L = \frac{1}{N} \sum_{i=1}^N \sum_{j \neq y_i} \max(0, f(x_i; W)_j - f(x_i; W)_{y_i} + 1) + \lambda R(W)$$

In common use:

L2 regularization

$$R(W) = \sum_k \sum_l W_{k,l}^2$$

L1 regularization

$$R(W) = \sum_k \sum_l |W_{k,l}|$$

Elastic net (L1 + L2)

$$R(W) = \sum_k \sum_l \beta W_{k,l}^2 + |W_{k,l}|$$

Max norm regularization (might see later)

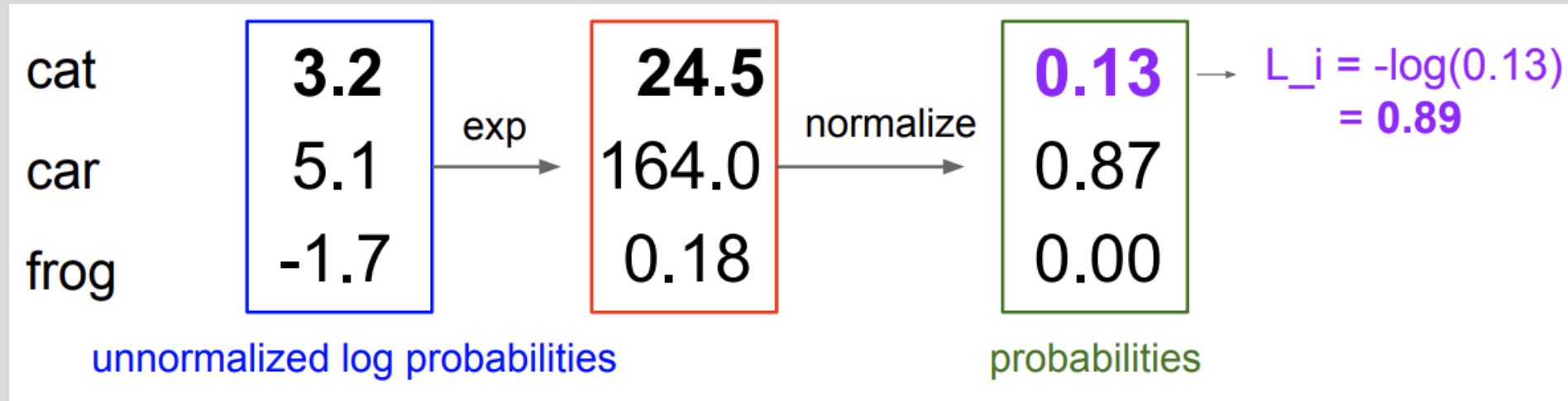
Dropout (will see later)

Fancier: Batch normalization, stochastic depth

Multiclass loss function

Softmax classifier(Multinomial Logistic Regression)

$$L_i = -\log\left(\frac{e^{sy_i}}{\sum_j e^{s_j}}\right) \quad L_i \Rightarrow [0, \infty)$$



Next time:

Optimization, Backpropagation, Neural Networks